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Polyphase power amplifier architecture for load insensitivity

Abstract

Polyphase power amplifiers for load insensitivity are disclosed. In certain embodiments, a polyphase transmit system includes an intermediate frequency transceiver including a first complex mixer that outputs a plurality of intermediate frequency transmit signals of different phases, and an intermediate frequency to radio frequency module including a second complex mixer that generates a plurality of radio frequency transmit signals of different phases based on the plurality of intermediate frequency transmit signals, and a polyphase power amplifier that receives the plurality of radio frequency transmit signals and outputs an amplified radio frequency signal. The polyphase transmit system further includes an antenna that transmits the amplified radio frequency signal.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority under 35 U.S.C. § 119 of U.S. Provisional Patent Application No. 63/266,090, filed Dec. 28, 2021 and titled “POLYPHASE POWER AMPLIFIER FOR LOAD INSENSITIVITY,” which is herein incorporated by reference in its entirety.

BACKGROUND

Field

(1) Embodiments of the invention relate to electronic systems, and in particular, to radio frequency (RF) electronics.

Description of the Related Technology

(2) Radio frequency (RF) communication systems wirelessly communicate RF signals using antennas.

(3) Examples of RF communication systems that utilize antennas for communication include, but are not limited to mobile phones, tablets, base stations, network access points, laptops, and wearable electronics. RF signals have a frequency in the range from about 30 kHz to 300 GHz, for instance, in the range of about 400 MHz to about 7.125 GHz for Frequency Range 1 (FR1) of the Fifth Generation (5G) communication standard or in the range of about 24.250 GHz to about 71.000 GHz for Frequency Range 2 (FR2) of the 5G communication standard.

SUMMARY

(4) In certain embodiments, the present disclosure relates to a mobile device. The mobile device includes a baseband processor configured to generate digital transmit data representing a transmit signal, and an intermediate frequency transceiver including a plurality of digital-to-analog converters configured to convert the digital transmit data into a plurality of analog baseband transmit signals of different phases, and a first complex mixer configured to generate a plurality of intermediate frequency transmit signals of different phases based on the plurality of analog baseband transmit signals. The mobile device further includes a front-end system including a second complex mixer configured to generate a plurality of radio frequency transmit signals of different phases based on the plurality of intermediate frequency transmit signals, and a polyphase power amplifier configured to receive the plurality of radio frequency transmit signals and to output an amplified radio frequency signal. The mobile device further includes an antenna configured to transmit the amplified radio frequency signal.

(5) In various embodiments, the polyphase power amplifier provides insensitivity to a variation in an impedance of the antenna. According to a number of embodiments, the variation is a change in voltage standing wave ratio (VSWR).

(6) In several embodiments, the plurality of intermediate frequency transmit signals includes an intermediate frequency in-phase transmit signal and an intermediate frequency quadrature-phase transmit signal, and the plurality of radio frequency transmit signals includes a radio frequency in-phase transmit signal and a radio frequency quadrature-phase transmit signal. According to a number of embodiments, the front-end system further includes a coarse phase shifter configured to selectively provide a ninety degree phase shift to the radio frequency in-phase transmit signal and the radio frequency quadrature-phase transmit signal. In accordance with various embodiments, the front-end system further includes a coarse phase shifter configured to provide a fine phase shift to the radio frequency in-phase transmit signal and the radio frequency quadrature-phase transmit signal. According to some embodiments, the front-end system further includes a polyphase filter configured to filter the radio frequency in-phase transmit signal and the radio frequency quadrature-phase transmit signal. In accordance with a number of embodiments, the polyphase amplifier includes a first controllable gain power amplifier configured to receive the radio

frequency in-phase transmit signal, a second controllable gain power amplifier configured to receive the radio frequency quadrature-phase transmit signal, and a combiner configured to generate the amplified radio frequency signal by combining an output of the first controllable gain power amplifier and an output of the second controllable gain power amplifier.

(7) In some embodiments, the antenna is a patch antenna.

(8) In various embodiments, the intermediate frequency transceiver includes a first local oscillator configured to provide the first complex mixer with a first plurality of local oscillator signals, and the front end system further includes a second local oscillator configured to provide the second complex mixer with a second plurality of local oscillator signals.

(9) In certain embodiments, the present disclosure relates to a polyphase transmit system. The polyphase transmit system includes an intermediate frequency transceiver including a first complex mixer configured to output a plurality of intermediate frequency transmit signals of different phases. The polyphase transmit system further includes an intermediate frequency to radio frequency module including a second complex mixer configured to generate a plurality of radio frequency transmit signals of different phases based on the plurality of intermediate frequency transmit signals, and a polyphase power amplifier configured to receive the plurality of radio frequency transmit signals and to output an amplified radio frequency signal. The polyphase transmit system further includes an antenna configured to transmit the amplified radio frequency signal.

(10) In several embodiments, the polyphase power amplifier provides insensitivity to a variation in an impedance of the antenna.

(11) In some embodiments, the plurality of intermediate frequency transmit signals includes an intermediate frequency in-phase transmit signal and an intermediate frequency quadrature-phase transmit signal, and the plurality of radio frequency transmit signals includes a radio frequency in-phase transmit signal and a radio frequency quadrature-phase transmit signal. According to a number of embodiments, the intermediate frequency to radio frequency module further includes a coarse phase shifter configured to selectively provide a ninety degree phase shift to the radio frequency in-phase transmit signal and the radio frequency quadrature-phase transmit signal. In accordance with various embodiments, the intermediate frequency to radio frequency module further includes a coarse phase shifter configured to provide a fine phase shift to the radio frequency in-phase transmit signal and the radio frequency quadrature-phase transmit signal. According to several embodiments, the intermediate frequency to radio frequency module further includes a polyphase filter configured to filter the radio frequency in-phase transmit signal and the radio frequency quadrature-phase transmit signal. In accordance with a number of embodiments, the polyphase amplifier includes a first controllable gain power amplifier configured to receive the radio frequency in-phase transmit signal, a second controllable gain power amplifier configured to receive the radio frequency quadrature-phase transmit signal, and a combiner configured to combine an output of the first controllable gain power amplifier and an output of the second controllable gain power amplifier.

(12) In several embodiments, the antenna is a patch antenna.

(13) In various embodiments, the intermediate frequency transceiver includes a first local oscillator configured to provide the first complex mixer with a first plurality of local oscillator signals, and the intermediate frequency to radio frequency module further includes a second local oscillator configured to provide the second complex mixer with a second plurality of local oscillator signals.

(14) In certain embodiments, the present disclosure relates to a method of transmitting in a mobile device. The method includes outputting a plurality of intermediate frequency transmit signals of different phases from a first complex mixer of an intermediate frequency transceiver, generating a plurality of radio frequency transmit signals of different phases based on the plurality of intermediate frequency transmit signals using a second complex mixer of an intermediate frequency to radio frequency module, amplifying the plurality of radio frequency transmit signals

to generate an amplified radio frequency signal using a polyphase power amplifier, and transmitting the amplified radio frequency signal using an antenna.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram of one example of a communication network.
 - (2) FIG. 2A is a schematic diagram of one example of a communication link using carrier aggregation.
 - (3) FIG. 2B illustrates various examples of uplink carrier aggregation for the communication link of FIG. 2A.
 - (4) FIG. 2C illustrates various examples of downlink carrier aggregation for the communication link of FIG. 2A.
 - (5) FIG. 3A is a schematic diagram of one example of a downlink channel using multi-input and multi-output (MIMO) communications.
 - (6) FIG. 3B is schematic diagram of one example of an uplink channel using MIMO communications.
 - (7) FIG. 3C is schematic diagram of another example of an uplink channel using MIMO communications.
 - (8) FIG. 4A is a schematic diagram of one example of a communication system that operates with beamforming.
 - (9) FIG. 4B is a schematic diagram of one example of beamforming to provide a transmit beam.
 - (10) FIG. 4C is a schematic diagram of one example of beamforming to provide a receive beam.
 - (11) FIG. 5A is a schematic diagram of one embodiment of a polyphase transmit system.
 - (12) FIG. 5B is a schematic diagram of another embodiment of a polyphase transmit system.
 - (13) FIG. 5C is a schematic diagram of one embodiment of mixers for a polyphase transmit system.
 - (14) FIG. 5D is a schematic diagram of a quadrature power amplifier for a polyphase transmit system.
 - (15) FIG. 5E is a first example of phase shifting for a polyphase transmit system.
 - (16) FIG. 5F is a second example of phase shifting for a polyphase transmit system.
 - (17) FIG. 5G is a third example of phase shifting for a polyphase transmit system.
 - (18) FIG. 5H is a fourth example of phase shifting for a polyphase transmit system.
 - (19) FIG. 5I is a fifth example of phase shifting for a polyphase transmit system.
 - (20) FIG. 5J is a schematic diagram of one embodiment of a polyphase filter for a polyphase transmit system.
 - (21) FIG. 5K is one example of gain versus frequency for a polyphase filter.
 - (22) FIG. 5L is one example of an in-phase/quadrature-phase (I/Q) diagram for frequency shifting in a polyphase transmit system.
 - (23) FIG. 5M is a schematic diagram of one embodiment of a polyphase transmit system using multiple antenna polarizations.
 - (24) FIG. 6A is a perspective view of one embodiment of a module that operates with beamforming.
 - (25) FIG. 6B is a cross-section of the module of FIG. 6A taken along the lines 6B-6B.
 - (26) FIG. 7 is a schematic diagram of one embodiment of a mobile device.
 - (27) FIG. 8 is a schematic diagram of a power amplifier system according to another embodiment.
- ### DETAILED DESCRIPTION OF EMBODIMENTS
- (28) The following detailed description of certain embodiments presents various descriptions of specific embodiments. However, the innovations described herein can be embodied in a multitude of different ways, for example, as defined and covered by the claims. In this description, reference

is made to the drawings where like reference numerals can indicate identical or functionally similar elements. It will be understood that elements illustrated in the figures are not necessarily drawn to scale. Moreover, it will be understood that certain embodiments can include more elements than illustrated in a drawing and/or a subset of the elements illustrated in a drawing. Further, some embodiments can incorporate any suitable combination of features from two or more drawings.

(29) The International Telecommunication Union (ITU) is a specialized agency of the United Nations (UN) responsible for global issues concerning information and communication technologies, including the shared global use of radio spectrum.

(30) The 3rd Generation Partnership Project (3GPP) is a collaboration between groups of telecommunications standard bodies across the world, such as the Association of Radio Industries and Businesses (ARIB), the Telecommunications Technology Committee (TTC), the China Communications Standards Association (CCSA), the Alliance for Telecommunications Industry Solutions (ATIS), the Telecommunications Technology Association (TTA), the European Telecommunications Standards Institute (ETSI), and the Telecommunications Standards Development Society, India (TSDSI).

(31) Working within the scope of the ITU, 3GPP develops and maintains technical specifications for a variety of mobile communication technologies, including, for example, second generation (2G) technology (for instance, Global System for Mobile Communications (GSM) and Enhanced Data Rates for GSM Evolution (EDGE)), third generation (3G) technology (for instance, Universal Mobile Telecommunications System (UMTS) and High Speed Packet Access (HSPA)), and fourth generation (4G) technology (for instance, Long Term Evolution (LTE) and LTE-Advanced).

(32) The technical specifications controlled by 3GPP can be expanded and revised by specification releases, which can span multiple years and specify a breadth of new features and evolutions.

(33) In one example, 3GPP introduced carrier aggregation (CA) for LTE in Release 10. Although initially introduced with two downlink carriers, 3GPP expanded carrier aggregation in Release 14 to include up to five downlink carriers and up to three uplink carriers. Other examples of new features and evolutions provided by 3GPP releases include, but are not limited to, License Assisted Access (LAA), enhanced LAA (eLAA), Narrowband Internet of things (NB-IOT), Vehicle-to-Everything (V2X), and High Power User Equipment (HPUE).

(34) 3GPP introduced Phase 1 of fifth generation (5G) technology in Release 15, and introduced Phase 2 of 5G technology in Release 16. Subsequent 3GPP releases will further evolve and expand 5G technology. 5G technology is also referred to herein as 5G New Radio (NR).

(35) 5G NR supports or plans to support a variety of features, such as communications over millimeter wave spectrum, beamforming capability, high spectral efficiency waveforms, low latency communications, multiple radio numerology, and/or non-orthogonal multiple access (NOMA). Although such RF functionalities offer flexibility to networks and enhance user data rates, supporting such features can pose a number of technical challenges.

(36) The teachings herein are applicable to a wide variety of communication systems, including, but not limited to, communication systems using advanced cellular technologies, such as LTE-Advanced, LTE-Advanced Pro, and/or 5G NR.

(37) FIG. 1 is a schematic diagram of one example of a communication network **10**. The communication network **10** includes a macro cell base station **1**, a small cell base station **3**, and various examples of user equipment (UE), including a first mobile device **2a**, a wireless-connected car **2b**, a laptop **2c**, a stationary wireless device **2d**, a wireless-connected train **2e**, a second mobile device **2f**, and a third mobile device **2g**.

(38) Although specific examples of base stations and user equipment are illustrated in FIG. 1, a communication network can include base stations and user equipment of a wide variety of types and/or numbers.

(39) For instance, in the example shown, the communication network **10** includes the macro cell base station **1** and the small cell base station **3**. The small cell base station **3** can operate with

relatively lower power, shorter range, and/or with fewer concurrent users relative to the macro cell base station **1**. The small cell base station **3** can also be referred to as a femtocell, a picocell, or a microcell. Although the communication network **10** is illustrated as including two base stations, the communication network **10** can be implemented to include more or fewer base stations and/or base stations of other types.

(40) Although various examples of user equipment are shown, the teachings herein are applicable to a wide variety of user equipment, including, but not limited to, mobile phones, tablets, laptops, IoT devices, wearable electronics, customer premises equipment (CPE), wireless-connected vehicles, wireless relays, and/or a wide variety of other communication devices. Furthermore, user equipment includes not only currently available communication devices that operate in a cellular network, but also subsequently developed communication devices that will be readily implementable with the inventive systems, processes, methods, and devices as described and claimed herein.

(41) The illustrated communication network **10** of FIG. **1** supports communications using a variety of cellular technologies, including, for example, 4G LTE and 5G NR. In certain implementations, the communication network **10** is further adapted to provide a wireless local area network (WLAN), such as WiFi. Although various examples of communication technologies have been provided, the communication network **10** can be adapted to support a wide variety of communication technologies.

(42) Various communication links of the communication network **10** have been depicted in FIG. **1**. The communication links can be duplexed in a wide variety of ways, including, for example, using frequency-division duplexing (FDD) and/or time-division duplexing (TDD). FDD is a type of radio frequency communications that uses different frequencies for transmitting and receiving signals. FDD can provide a number of advantages, such as high data rates and low latency. In contrast, TDD is a type of radio frequency communications that uses about the same frequency for transmitting and receiving signals, and in which transmit and receive communications are switched in time. TDD can provide a number of advantages, such as efficient use of spectrum and variable allocation of throughput between transmit and receive directions.

(43) In certain implementations, user equipment can communicate with a base station using one or more of 4G LTE, 5G NR, and WiFi technologies. In certain implementations, enhanced license assisted access (eLAA) is used to aggregate one or more licensed frequency carriers (for instance, licensed 4G LTE and/or 5G NR frequencies), with one or more unlicensed carriers (for instance, unlicensed WiFi frequencies).

(44) As shown in FIG. **1**, the communication links include not only communication links between UE and base stations, but also UE to UE communications and base station to base station communications. For example, the communication network **10** can be implemented to support self-fronthaul and/or self-backhaul (for instance, as between mobile device **2g** and mobile device **20**).

(45) The communication links can operate over a wide variety of frequencies. In certain implementations, communications are supported using 5G NR technology over one or more frequency bands that are less than 6 Gigahertz (GHz) and/or over one or more frequency bands that are greater than 6 GHz. For example, the communication links can serve Frequency Range 1 (FR1), Frequency Range 2 (FR2), or a combination thereof. In one embodiment, one or more of the mobile devices support a HPUE power class specification.

(46) In certain implementations, a base station and/or user equipment communicates using beamforming. For example, beamforming can be used to focus signal strength to overcome path losses, such as high loss associated with communicating over high signal frequencies. In certain embodiments, user equipment, such as one or more mobile phones, communicate using beamforming on millimeter wave frequency bands which include 5G FR2. Thus, as used herein a millimeter wave signal can include traditional millimeter waves (30 GHz to 300 GHz) as well as upper centimeter wave frequencies in the range of 24 GHz to 30 GHz. Cellular user equipment can

communicate using beamforming and/or other techniques over a wide range of frequencies, including, for example, FR2-1 (24 GHz to 52 GHz), FR2-2 (52 GHz to 71 GHz), and/or FR1 (400 MHz to 7125 MHz).

(47) Different users of the communication network **10** can share available network resources, such as available frequency spectrum, in a wide variety of ways.

(48) In one example, frequency division multiple access (FDMA) is used to divide a frequency band into multiple frequency carriers. Additionally, one or more carriers are allocated to a particular user. Examples of FDMA include, but are not limited to, single carrier FDMA (SC-FDMA) and orthogonal FDMA (OFDMA). OFDMA is a multicarrier technology that subdivides the available bandwidth into multiple mutually orthogonal narrowband subcarriers, which can be separately assigned to different users.

(49) Other examples of shared access include, but are not limited to, time division multiple access (TDMA) in which a user is allocated particular time slots for using a frequency resource, code division multiple access (CDMA) in which a frequency resource is shared amongst different users by assigning each user a unique code, space-divisional multiple access (SDMA) in which beamforming is used to provide shared access by spatial division, and non-orthogonal multiple access (NOMA) in which the power domain is used for multiple access. For example, NOMA can be used to serve multiple users at the same frequency, time, and/or code, but with different power levels.

(50) Enhanced mobile broadband (eMBB) refers to technology for growing system capacity of LTE networks. For example, eMBB can refer to communications with a peak data rate of at least 10 Gbps and a minimum of 100 Mbps for each user. Ultra-reliable low latency communications (uRLLC) refers to technology for communication with very low latency, for instance, less than 2 milliseconds. uRLLC can be used for mission-critical communications such as for autonomous driving and/or remote surgery applications. Massive machine-type communications (mMTC) refers to low cost and low data rate communications associated with wireless connections to everyday objects, such as those associated with Internet of Things (IoT) applications.

(51) The communication network **10** of FIG. **1** can be used to support a wide variety of advanced communication features, including, but not limited to, eMBB, uRLLC, and/or mMTC.

(52) FIG. **2A** is a schematic diagram of one example of a communication link using carrier aggregation. Carrier aggregation can be used to widen bandwidth of the communication link by supporting communications over multiple frequency carriers, thereby increasing user data rates and enhancing network capacity by utilizing fragmented spectrum allocations.

(53) In the illustrated example, the communication link is provided between a base station **21** and a mobile device **22**. As shown in FIG. **2A**, the communications link includes a downlink channel used for RF communications from the base station **21** to the mobile device **22**, and an uplink channel used for RF communications from the mobile device **22** to the base station **21**.

(54) Although FIG. **2A** illustrates carrier aggregation in the context of FDD communications, carrier aggregation can also be used for TDD communications.

(55) In certain implementations, a communication link can provide asymmetrical data rates for a downlink channel and an uplink channel. For example, a communication link can be used to support a relatively high downlink data rate to enable high speed streaming of multimedia content to a mobile device, while providing a relatively slower data rate for uploading data from the mobile device to the cloud.

(56) In the illustrated example, the base station **21** and the mobile device **22** communicate via carrier aggregation, which can be used to selectively increase bandwidth of the communication link. Carrier aggregation includes contiguous aggregation, in which contiguous carriers within the same operating frequency band are aggregated. Carrier aggregation can also be non-contiguous, and can include carriers separated in frequency within a common band or in different bands.

(57) In the example shown in FIG. **2A**, the uplink channel includes three aggregated component

carriers f.sub.UL1, f.sub.UL2, and f.sub.UL3. Additionally, the downlink channel includes five aggregated component carriers f.sub.DL1, f.sub.DL2, f.sub.DL3, f.sub.DL4, and f.sub.DL5.

Although one example of component carrier aggregation is shown, more or fewer carriers can be aggregated for uplink and/or downlink. Moreover, a number of aggregated carriers can be varied over time to achieve desired uplink and downlink data rates.

(58) For example, a number of aggregated carriers for uplink and/or downlink communications with respect to a particular mobile device can change over time. For example, the number of aggregated carriers can change as the device moves through the communication network and/or as network usage changes over time.

(59) FIG. 2B illustrates various examples of uplink carrier aggregation for the communication link of FIG. 2A. FIG. 2B includes a first carrier aggregation scenario **31**, a second carrier aggregation scenario **32**, and a third carrier aggregation scenario **33**, which schematically depict three types of carrier aggregation.

(60) The carrier aggregation scenarios **31-33** illustrate different spectrum allocations for a first component carrier full, a second component carrier f.sub.UL2, and a third component carrier f.sub.UL3. Although FIG. 2B is illustrated in the context of aggregating three component carriers, carrier aggregation can be used to aggregate more or fewer carriers. Moreover, although illustrated in the context of uplink, the aggregation scenarios are also applicable to downlink.

(61) The first carrier aggregation scenario **31** illustrates intra-band contiguous carrier aggregation, in which component carriers that are adjacent in frequency and in a common frequency band are aggregated. For example, the first carrier aggregation scenario **31** depicts aggregation of component carriers f.sub.UL1, f.sub.UL2, and f.sub.UL3 that are contiguous and located within a first frequency band BAND1.

(62) With continuing reference to FIG. 2B, the second carrier aggregation scenario **32** illustrates intra-band non-continuous carrier aggregation, in which two or more components carriers that are non-adjacent in frequency and within a common frequency band are aggregated. For example, the second carrier aggregation scenario **32** depicts aggregation of component carriers f.sub.UL1, f.sub.UL2, and f.sub.UL3 that are non-contiguous, but located within a first frequency band BAND1.

(63) The third carrier aggregation scenario **33** illustrates inter-band non-contiguous carrier aggregation, in which component carriers that are non-adjacent in frequency and in multiple frequency bands are aggregated. For example, the third carrier aggregation scenario **33** depicts aggregation of component carriers f.sub.UL1 and f.sub.UL2 of a first frequency band BAND1 with component carrier f.sub.UL3 of a second frequency band BAND2.

(64) FIG. 2C illustrates various examples of downlink carrier aggregation for the communication link of FIG. 2A. The examples depict various carrier aggregation scenarios **34-38** for different spectrum allocations of a first component carrier f.sub.DL1, a second component carrier f.sub.DL2, a third component carrier f.sub.DL3, a fourth component carrier f.sub.DL4, and a fifth component carrier f.sub.DL5. Although FIG. 2C is illustrated in the context of aggregating five component carriers, carrier aggregation can be used to aggregate more or fewer carriers. Moreover, although illustrated in the context of downlink, the aggregation scenarios are also applicable to uplink.

(65) The first carrier aggregation scenario **34** depicts aggregation of component carriers that are contiguous and located within the same frequency band. Additionally, the second carrier aggregation scenario **35** and the third carrier aggregation scenario **36** illustrates two examples of aggregation that are non-contiguous, but located within the same frequency band. Furthermore, the fourth carrier aggregation scenario **37** and the fifth carrier aggregation scenario **38** illustrates two examples of aggregation in which component carriers that are non-adjacent in frequency and in multiple frequency bands are aggregated. As a number of aggregated component carriers increases, a complexity of possible carrier aggregation scenarios also increases.

(66) With reference to FIGS. 2A-2C, the individual component carriers used in carrier aggregation

can be of a variety of frequencies, including, for example, frequency carriers in the same band or in multiple bands. Additionally, carrier aggregation is applicable to implementations in which the individual component carriers are of about the same bandwidth as well as to implementations in which the individual component carriers have different bandwidths.

(67) Certain communication networks allocate a particular user device with a primary component carrier (PCC) or anchor carrier for uplink and a PCC for downlink. Additionally, when the mobile device communicates using a single frequency carrier for uplink or downlink, the user device communicates using the PCC. To enhance bandwidth for uplink communications, the uplink PCC can be aggregated with one or more uplink secondary component carriers (SCCs). Additionally, to enhance bandwidth for downlink communications, the downlink PCC can be aggregated with one or more downlink SCCs.

(68) In certain implementations, a communication network provides a network cell for each component carrier. Additionally, a primary cell can operate using a PCC, while a secondary cell can operate using a SCC. The primary and secondary cells may have different coverage areas, for instance, due to differences in frequencies of carriers and/or network environment.

(69) License assisted access (LAA) refers to downlink carrier aggregation in which a licensed frequency carrier associated with a mobile operator is aggregated with a frequency carrier in unlicensed spectrum, such as WiFi. LAA employs a downlink PCC in the licensed spectrum that carries control and signaling information associated with the communication link, while unlicensed spectrum is aggregated for wider downlink bandwidth when available. LAA can operate with dynamic adjustment of secondary carriers to avoid WiFi users and/or to coexist with WiFi users. Enhanced license assisted access (eLAA) refers to an evolution of LAA that aggregates licensed and unlicensed spectrum for both downlink and uplink. Furthermore, NR-U can operate on top of LAA/eLAA over a 5 GHz band (5150 to 5925 MHz) and/or a 6 GHz band (5925 MHz to 7125 MHz).

(70) FIG. 3A is a schematic diagram of one example of a downlink channel using multi-input and multi-output (MIMO) communications. FIG. 3B is schematic diagram of one example of an uplink channel using MIMO communications.

(71) MIMO communications use multiple antennas for simultaneously communicating multiple data streams over common frequency spectrum. In certain implementations, the data streams operate with different reference signals to enhance data reception at the receiver. MIMO communications benefit from higher SNR, improved coding, and/or reduced signal interference due to spatial multiplexing differences of the radio environment.

(72) MIMO order refers to a number of separate data streams sent or received. For instance, MIMO order for downlink communications can be described by a number of transmit antennas of a base station and a number of receive antennas for UE, such as a mobile device. For example, two-by-two (2×2) DL MIMO refers to MIMO downlink communications using two base station antennas and two UE antennas. Additionally, four-by-four (4×4) DL MIMO refers to MIMO downlink communications using four base station antennas and four UE antennas.

(73) In the example shown in FIG. 3A, downlink MIMO communications are provided by transmitting using M antennas **43a**, **43b**, **43c**, . . . **43m** of the base station **41** and receiving using N antennas **44a**, **44b**, **44c**, . . . **44n** of the mobile device **42**. Accordingly, FIG. 3A illustrates an example of m×n DL MIMO.

(74) Likewise, MIMO order for uplink communications can be described by a number of transmit antennas of UE, such as a mobile device, and a number of receive antennas of a base station. For example, 2×2 UL MIMO refers to MIMO uplink communications using two UE antennas and two base station antennas. Additionally, 4×4 UL MIMO refers to MIMO uplink communications using four UE antennas and four base station antennas.

(75) In the example shown in FIG. 3B, uplink MIMO communications are provided by transmitting using N antennas **44a**, **44b**, **44c**, . . . **44n** of the mobile device **42** and receiving using M antennas

43a, 43b, 43c, . . . 43m of the base station **41**. Accordingly, FIG. 3B illustrates an example of $n \times m$ UL MIMO.

(76) By increasing the level or order of MIMO, bandwidth of an uplink channel and/or a downlink channel can be increased.

(77) MIMO communications are applicable to communication links of a variety of types, such as FDD communication links and TDD communication links.

(78) FIG. 3C is schematic diagram of another example of an uplink channel using MIMO communications. In the example shown in FIG. 3C, uplink MIMO communications are provided by transmitting using N antennas **44a, 44b, 44c, . . . 44n** of the mobile device **42**. Additionally a first portion of the uplink transmissions are received using M antennas **43a1, 43b1, 43c1, . . . 43m1** of a first base station **41a**, while a second portion of the uplink transmissions are received using M antennas **43a2, 43b2, 43c2, . . . 43m2** of a second base station **41b**. Additionally, the first base station **41a** and the second base station **41b** communicate with one another over wired, optical, and/or wireless links.

(79) The MIMO scenario of FIG. 3C illustrates an example in which multiple base stations cooperate to facilitate MIMO communications.

(80) FIG. 4A is a schematic diagram of one example of a communication system **110** that operates with beamforming. The communication system **110** includes a transceiver **105**, signal conditioning circuits **104a1, 104a2 . . . 104an, 104b1, 104b2 . . . 104bn, 104m1, 104m2 . . . 104mn**, and an antenna array **102** that includes antenna elements **103a1, 103a2 . . . 103an, 103b1, 103b2 . . . 103bn, 103m1, 103m2 . . . 103mn**.

(81) Communications systems that communicate using millimeter wave carriers can employ an antenna array to provide beam formation and directivity for transmission and/or reception of signals.

(82) For example, in the illustrated embodiment, the communication system **110** includes an array **102** of $m \times n$ antenna elements, which are each controlled by a separate signal conditioning circuit, in this embodiment. As indicated by the ellipses, the communication system **110** can be implemented with any suitable number of antenna elements and signal conditioning circuits.

(83) With respect to signal transmission, the signal conditioning circuits can provide transmit signals to the antenna array **102** such that signals radiated from the antenna elements combine using constructive and destructive interference to generate an aggregate transmit signal exhibiting beam-like qualities with more signal strength propagating in a given direction away from the antenna array **102**.

(84) In the context of signal reception, the signal conditioning circuits process the received signals (for instance, by separately controlling received signal phases) such that more signal energy is received when the signal is arriving at the antenna array **102** from a particular direction.

Accordingly, the communication system **110** also provides directivity for reception of signals.

(85) The relative concentration of signal energy into a transmit beam or a receive beam can be enhanced by increasing the size of the array. For example, with more signal energy focused into a transmit beam, the signal is able to propagate for a longer range while providing sufficient signal level for RF communications. For instance, a signal with a large proportion of signal energy focused into the transmit beam can exhibit high effective isotropic radiated power (EIRP).

(86) In the illustrated embodiment, the transceiver **105** provides transmit signals to the signal conditioning circuits and processes signals received from the signal conditioning circuits. As shown in FIG. 4A, the transceiver **105** generates control signals for the signal conditioning circuits. The control signals can be used for a variety of functions, such as controlling the gain and phase of transmitted and/or received signals to control beamforming.

(87) FIG. 4B is a schematic diagram of one example of beamforming to provide a transmit beam. FIG. 4B illustrates a portion of a communication system including a first signal conditioning circuit **114a**, a second signal conditioning circuit **114b**, a first antenna element **113a**, and a second antenna

element **113b**.

(88) Although illustrated as included two antenna elements and two signal conditioning circuits, a communication system can include additional antenna elements and/or signal conditioning circuits. For example, FIG. 4B illustrates one embodiment of a portion of the communication system **110** of FIG. 4A.

(89) The first signal conditioning circuit **114a** includes a first phase shifter **130a**, a first power amplifier **131a**, a first low noise amplifier (LNA) **132a**, and switches for controlling selection of the power amplifier **131a** or LNA **132a**. Additionally, the second signal conditioning circuit **114b** includes a second phase shifter **130b**, a second power amplifier **131b**, a second LNA **132b**, and switches for controlling selection of the power amplifier **131b** or LNA **132b**.

(90) Although one embodiment of signal conditioning circuits is shown, other implementations of signal conditioning circuits are possible. For instance, in one example, a signal conditioning circuit includes one or more band filters, duplexers, and/or other components.

(91) In the illustrated embodiment, the first antenna element **113a** and the second antenna element **113b** are separated by a distance d . Additionally, FIG. 4B has been annotated with an angle θ , which in this example has a value of about 90° when the transmit beam direction is substantially perpendicular to a plane of the antenna array and a value of about 0° when the transmit beam direction is substantially parallel to the plane of the antenna array.

(92) By controlling the relative phase of the transmit signals provided to the antenna elements **113a**, **113b**, a desired transmit beam angle θ can be achieved. For example, when the first phase shifter **130a** has a reference value of 0° , the second phase shifter **130b** can be controlled to provide a phase shift of about $-2\pi f(d/v)\cos \theta$ radians, where f is the fundamental frequency of the transmit signal, d is the distance between the antenna elements, v is the velocity of the radiated wave, and π is the mathematic constant pi.

(93) In certain implementations, the distance d is implemented to be about $\frac{1}{2}\lambda$, where λ is the wavelength of the fundamental component of the transmit signal. In such implementations, the second phase shifter **130b** can be controlled to provide a phase shift of about $-\pi \cos \theta$ radians to achieve a transmit beam angle θ .

(94) Accordingly, the relative phase of the phase shifters **130a**, **130b** can be controlled to provide transmit beamforming. In certain implementations, a baseband processor and/or a transceiver (for example, the transceiver **105** of FIG. 4A) controls phase values of one or more phase shifters and gain values of one or more controllable amplifiers to control beamforming.

(95) FIG. 4C is a schematic diagram of one example of beamforming to provide a receive beam. FIG. 4C is similar to FIG. 4B, except that FIG. 4C illustrates beamforming in the context of a receive beam rather than a transmit beam.

(96) As shown in FIG. 4C, a relative phase difference between the first phase shifter **130a** and the second phase shifter **130b** can be selected to about equal to $-2\pi f(d/v)\cos \theta$ radians to achieve a desired receive beam angle θ . In implementations in which the distance d corresponds to about $\frac{1}{2}\lambda$, the phase difference can be selected to about equal to $-\pi \cos \theta$ radians to achieve a receive beam angle θ .

(97) Although various equations for phase values to provide beamforming have been provided, other phase selection values are possible, such as phase values selected based on implementation of an antenna array, implementation of signal conditioning circuits, and/or a radio environment.

(98) FIG. 5A is a schematic diagram of one embodiment of a polyphase transmit system **210**. The polyphase transmit system **210** includes a polyphase generator **201**, a polyphase power amplifier **202**, and an antenna **203**.

(99) As shown in FIG. 5A, the polyphase generator **201** receives a transmit signal TX, which the polyphase generator **201** processes to generate polyphase transmit signals $\phi_1, \phi_2, \dots \phi_{\text{sub.n}}$ (collectively referred to as RF.sub.TX) of different phases. The polyphase power amplifier **202** processes RF.sub.TX to generate an amplified RF transmit signal for transmission on the antenna

203.

(100) Although shown as amplifying of transmit signals $\phi_{\text{sub.1}}$, $\phi_{\text{sub.2}}$, . . . $\phi_{\text{sub.n}}$ of three different phases, the teachings herein are applicable to polyphase transmit systems using additional phases. For example, in certain implementations, a polyphase transmit systems operates on four or more transmit signal phases. For instance, when using four transmit signal phases, the transmit signals can have phases of 0° , 90° , 180° , and 270° in some implementations.

(101) The polyphase transmit system **210** of FIG. 5A depicts a full polyphase power amplifier architecture, which provides load insensitivity, for instance, insensitivity to changes in voltage standing wave ratio (VSWR) of the antenna **203**.

(102) In certain implementations, the polyphase generator **201** serves to provide a full quadrature lineup (with phases of 0° , 90° , 180° , and 270°) to intermediate frequency/local oscillator (IF/LO) from input to output.

(103) The number of phases n processed by the polyphase power amplifier **202** can be any suitable number. For example, the number of phases n can be 3 or more, or more preferably, 4 or more. For instance, in certain embodiments, the phases includes an in-phase signal (I or 0°), an inverted in-phase signal (I_B or 180°), a quadrature phase signal (Q or 90°), and an inverted quadrature phase signal (Q_B or 270°).

(104) In certain implementations, the transmit signal TX corresponds to digital data from a baseband processor, which is upconverted by a first complex mixer of the polyphase generator **201** to intermediate frequency (IF). Additionally, multiple IF signals (for example, 0° , 90° , 180° , and 270° IF signals) are maintained. The IF signals are upconverted to RF by a second complex mixer of the polyphase generator **201**, and multiple RF signals (for example, 0° , 90° , 180° , and 270° RF signals) are maintained.

(105) By using multiple IF and RF signals in this manner, phase shifting can be performed at IF and/or RF. For example, 90 degree phase shifts can be easily achieved by signal selection in the IF or RF domains, while when combined with power control with each path further phase states can be created by summing two of the phases with a different coefficient (set by power control).

(106) In certain implementations, the carrier frequency of the RF signal transmitted by a polyphase transmit system corresponds to a 5G frequency band in FR2. Table 1 below depicts various examples of such 5G frequency bands in FR2. Although various examples of frequency bands have been described, other examples are possible.

(107) TABLE-US-00001

TABLE 1 5G Frequency Band Duplex		UL/DL Low	UL/DL High
Type	[MHz]	[MHz]	
n257 TDD	26500	29500	n258 TDD 24250 27500
n259 TDD	39500	43500	
n260 TDD	37000	40000	n261 TDD 27500 28350
n262 TDD	47200	48200	n263 TDD 57000 71000

(108) FIG. 5B is a schematic diagram of another embodiment of a polyphase transmit system **250**. The polyphase transmit system **250** includes a digital baseband circuit **211** (also referred to as a baseband processor), an IF transceiver **212**, and an IF to RF module **213**, and an antenna array **214**.

(109) In the illustrated embodiment, the IF transceiver **212** includes an I-path digital-to-analog converter (IDAC) **221**, a Q-path digital-to-analog converter (QDAC) **222**, an IF local oscillator (IFLO) **230**, and a first group of mixers **231-234** (including a first mixer **231**, a second mixer **232**, a third mixer **233**, and a fourth mixer **234**, also collectively referred to as a first complex mixer **231-234**).

(110) The IF to RF module **213** includes an RF local oscillator (RFLO) **235**, a coarse phase shifter **236** (90 degree, in this example), a fine phase shifter **237** (45 degree when weighting coefficients are equal, in this example), a polyphase filter **238**, polyphase power amplifiers **239**, and a second group of mixers **251-254** (including a first mixer **251**, a second mixer **252**, a third mixer **253**, and a fourth mixer **254**, also collectively referred to as a second complex mixer **251-254**). In certain implementations, the IF to RF module **213** is included as part of an RF front end system of a mobile device.

(111) As shown in FIG. 5B, the antenna array **214** includes a first patch antenna **248a**, a second

patch antenna **248b**, a third patch antenna **248c**, and a fourth patch antenna **248d**, in this example. Although shown as including signal feed for each patch antenna, the teachings herein are applicable to patch antennas with differential drive as well as to patch antennas with feeds for multiple polarizations.

(112) In the illustrated embodiment, the polyphase power amplifiers **239** include first power amplifiers **241a-241d** (**241a**, **241b**, **241c**, and **241d**), second power amplifiers **242a-242d** (**242a**, **242b**, **242c**, and **242d**), RF combiners (implemented as 90 degree hybrids, in this example) **243a-243d** (**243a**, **243b**, **243c**, and **243d**), and termination resistors **244a-244d** (**244a**, **244b**, **244c**, and **244d**).

(113) The digital baseband circuit **211** provides digital data to the IF transceiver **212**. The digital data represents a transmit signal to be transmitted by the polyphase transmit system **250**. The IDAC **221** and the QDAC **222** generate an analog I baseband signal and an analog Q baseband signal that are provided to the first complex mixer **231-234** for upconversion to IF. The analog I baseband signal and the analog Q baseband signal have a 90 degree or quadrature phase relationship. In certain implementations, both the analog I baseband signal and the analog Q baseband signal are differential signals, and thus four analog baseband signal phases (0, 90, 180, and 270 degrees) are provided to the first complex mixer **231-234**.

(114) The IFLO **230** provides an I clock signal to the first mixer **231** and the second mixer **232**, and a Q clock signal to the third mixer **233** and the fourth mixer **234**. The I clock signal and the Q clock signal have a quadrature phase relationship. In certain implementations, both the I clock signal and the Q clock signal are differential signals.

(115) With continuing reference to FIG. 5B, the IF transceiver **212** provides an IF I signal and an IF Q signal to the IF to RF module **213** over an IF interface. In certain implementations, both the IF I signal and the IF Q signal are differential signals, and thus the IF to RF module **213** receives four IF signal phases (0, 90, 180, and 270 degrees).

(116) Although one IF to RF module **213** is shown, in certain embodiments multiple IF to RF modules are present, and each IF to RF module can receive the same or different pair of IF I and IF Q signals from the IF transceiver **212**. In one embodiment, the IF transceiver **212** provides the same IF I signal and IF Q signal to each of the IF to RF modules. In another embodiment, the IF transceiver **212** provides coarse phase shifting (for example, using the coarse phase shifting configurations discussed further below) to generate multiple phase-shifted versions of the IF I signal and IF Q signal. Additionally, each IF to RF module receives an IF I signal and an IF Q signal with a desired coarse phase shift.

(117) The second group of mixers **251-254** generates an RF I signal and an RF Q signal based on frequency shifting the IF I signal and the IF Q signal using an in-phase local oscillator (ILO) signal and a quadrature-phase local oscillator (QLO) signal from the RFLO **235**. The ILO signal and the QLO signal have a quadrature phase relationship. In certain implementations, the ILO signal and the QLO signal are differential signals.

(118) The RF I signal and the RF Q signal are provided to the coarse phase shifter **236** for coarse phase shifting. In certain implementations, the RF I signal and the RF Q signal are differential signals, and thus the coarse phase shifter **236** receives four phases of RF signal (0, 90, 180, and 270 degrees).

(119) As shown in FIG. 5B, the fine phase shifter **237** further phase shifts the RF I signal and the RF Q signal after coarse phase shifting, and the polyphase filter **238** filters the RF I signal and the RF Q signal after fine phase shifting. In certain implementations, the RF I signal and the RF Q signal each remain as differential signals all the way up to the polyphase power amplifiers **239**.

(120) In certain implementations, one coarse phase shifter, one fine phase shifter, and one polyphase filter is included for each pair of power amplifiers associated with a given patch antenna. Thus, four sets of coarse phase shifters, fine phase shifters, and polyphase filters (with each set referred to as a phase shifting and filtering channel) can be included for the depicted polyphase

power amplifiers **239** of FIG. 5B. Such phase shifting and filtering channels can receive the same or different RF I signal and RF Q signal as inputs. In one example, each of the phase shifting and filtering channels receives the same copy of the RF I signal and the RF Q signal from the second complex mixer **251-254**.

(121) As shown in FIG. 5B, the polyphase power amplifiers **239** amplify and combine the RF I and RF Q signals to generate an RF transmit signal that is transmitted on a respective patch antenna of the antenna array **214**. Although shown as including four antennas, more or fewer antennas (and corresponding power amplifiers) can be included to achieve a desired number of RF signal channels. For example, the polyphase transmit system **250** can be used for beamforming a transmit beam, and the number of signal channels can be chosen based on desired beamforming characteristics needed for a particular design or application.

(122) The polyphase transmit system **250** of FIG. 5B includes quadrature coupled power amplifiers for antenna load insensitivity. For example, two power amplifiers coupled in quadrature are less sensitive to the load, which is a significant issue for antenna arrays used in beamforming applications as one transmit path is loaded by the other transmit paths by ways of coupling of the antenna patches. Such a load may depend on the beam steering, since loading is dependent on signal phase. By using quadrature power amplifiers, load insensitivity is improved.

(123) Moreover, using a full polyphase design (with 0, 90, 180, and 270 degree signals available along the path), 90 degree phase shifts are provided by selection of the appropriate signal phase (virtually free from a hardware and design cost perspective). Furthermore, with quadrature IQ and LO/RF image in up conversion is also eliminated. In addition to no image, selection of LO+IF or LO-IF to achieve frequency shifting is also enabled.

(124) Although shown for the case of a 90 degree coarse phase shift and a 45 degree fine phase shift, the polyphase transmit system **250** of FIG. 5B also provides other phase shifts using the phase shifters. For example, with respect to the fine phase shifter **237**, when using RF I and RF Q of equal weighting, $0.707 \cdot I + 0.707 \cdot Q$ or 45 degree phase shifting can be achieved. However, by varying a gain of the RF I and RF Q signals (through gain control with a desired gain coefficient for each signal path through the fine phase shifter **237**), other fine phase shifts can be achieved. Thus, the fine phase shifter **238** can weigh the RF I signal with a first controllable coefficient and weight the RF Q signal with a second controllable coefficient to achieve a desired amount of phase shifting.

(125) The polyphase filter **238** can be used to provide filtering can be used to provide frequency notches or other desired signal filtering to enhance performance of the polyphase transmit system **250**.

(126) As noted above, phase shifting can be performed at IF and/or RF. Furthermore, reversal of the phases from the IFLO **230** and/or RFLO **235** can be used to provide phase shifting using local oscillators, also referred to as performing phase shifting at LO.

(127) Thus, in FIG. 5B, one quad IF is depicted as being provided to one complex mixer and splitting different phases at RF. However, other implementations of phase shifting are possible.

(128) In one embodiment, one quad IF is provided to multiple complex mixers, and different phases at LO are used to generate multiple phase-shifted RF signals.

(129) In another embodiment, the IF is split into different phases using multiple complex mixers.

(130) In yet another embodiment, a mix of IF, LO, and/or RF phase shifting is used.

(131) Thus, when using full quadrature (0, 90, 180, and 270 degree) signal processing, phase shifts and frequency shifts at any point along the signal chain is seen at the RF transmit signal provided at the antenna.

(132) FIG. 5C is a schematic diagram of one embodiment of mixers **310** for a polyphase transmit system. The mixers **310** include a first mixer **301**, a second mixer **302**, a third mixer **303**, and a fourth mixer **304**. The arrangement of mixers **310** is also referred to herein as a complex mixer.

(133) The complex mixer **310** receives an I signal and a Q signal, each of which can be in

differential form and which are in quadrature with respect to one another. The first mixer **301** mixes the I signal and an LOI signal, which can also be differential. The second mixer **302** mixes the Q signal and the LOI signal. The third mixer **303** mixes the I signal and an LOQ signal, which can also be differential. The fourth mixer **304** mixes the Q signal and the LOQ signal.

(134) Signal addition **305** of the first mixer **301** and the third mixer **303** is used to generate an RF I signal, which can be differential. Signal subtraction **306** of the second mixer **302** and the fourth mixer **304** can be used to generate an RF Q signal, which can be differential.

(135) In one embodiment, all depicted signals are differential. Thus, 0, 90, 180, and 270 degree signals are available at all points for the RF, LO, and IF signals. Additionally, a sign change can be achieved by swapping the positive and negative component of a signal. Furthermore, 90 degree phase shifts are available. Additionally, swapping I and Q can provide positive or negative shifts. Moreover, frequency shifts ($RF=LO+IF$ or $RF=LO-IF$) can be achieved using the complex mixer **310**.

(136) In one example a 34 GHz LO and ± 6 GHz IF are used to provide 28 GHz or 40 GHz RF signals, thus providing coverage of two 5G FR2 frequency bands using a common transmitter. Thus, multi-band performance of two or more 5G FR2 frequency bands can be achieved using common components. In contrast, typical conventional phase shifting schemes are narrow band and cannot easily accommodate multiple frequency bands.

(137) FIG. 5D is a schematic diagram of a quadrature power amplifier **320** for a polyphase transmit system. The quadrature power amplifier **320** includes a controllable gain in-phase power amplifier (IPA) **311**, a controllable gain quadrature-phase power amplifier (QPA) **312**, a 90° hybrid **313**, and a termination resistor **314**. The controllable gain IPA **311** has a separately controllable gain from the controllable gain QPA **312**.

(138) The controllable gain IPA **311** includes an input that receives the RFI signal, and an output connected to a 0° port of the hybrid **313**. The termination resistor is connected to an isolation (ISO) port of the hybrid **313**. The controllable gain QPA **312** includes an input that receives the RFQ signal, and an output connected to a 0° port of the hybrid **313**. An antenna (for example, a patch antenna) is coupled an output (OUT) port of the hybrid **313** that provides a combined RF output signal (which is a combination of the amplified RF signals from the controllable gain IPA **311** and the controllable gain QPA **312**).

(139) The quadrature power amplifier **320** exhibits load insensitivity.

(140) FIGS. 5E-5I exhibit various examples of phase shifting for a polyphase transmit system. The phase shifting can occur at IF or RF. Furthermore, multiple instantiations of such phase shifting structures can be used to provide phase shifting at both IF and RF. The depicted phase shifters receive I (0°), I_B (180°), Q (90°), and Q_B (270°) signals, which are phase shifted to generate four phase-shifted output signals. As used herein, I_B is also referred to as $\bar{I}$, and Q_B is also referred to as $\bar{Q}$.

(141) In FIGS. 5E-5H, coarse phase shifting is provided using a crossbar switch. In FIG. 5I, fine phase shifting is achieved by using controllable gain in each signal path and combining the outputs of various signal paths to generate fine phase shifted signals.

(142) FIG. 5E is a first example of phase shifting for a polyphase transmit system. In FIG. 5E, the crossbar switch connects from I input to I output, from I_B input to I_B output, from Q input to Q output, and from Q_B input to Q_B output. Thus, 0° of phase shift is provided.

(143) FIG. 5F is a second example of phase shifting for a polyphase transmit system. In FIG. 5F, the crossbar switch connects from I input to Q output, from Q input to I_B output, from I_B input to Q_B output, and from Q_B input to I output. Thus, 90° of phase shift is provided.

(144) FIG. 5G is a third example of phase shifting for a polyphase transmit system. In FIG. 5G, the crossbar switch connects from I input to Q_B output, from Q input to I output, from I_B input to Q output, and from Q_B input to I_B output. Thus, -90° of phase shift is provided.

(145) FIG. 5H is a fourth example of phase shifting for a polyphase transmit system. In FIG. 5H,

the crossbar switch connects from I input to I_B output, from Q input to Q_B output, from I_B input to I output, and from Q_B input to Q output. Thus, 180° of phase shift is provided.

(146) FIG. 5I is a fifth example of phase shifting for a polyphase transmit system. In FIG. 5I, each of the I input, Q input, I_B input, and Q_B input are weighted with a 0.707 coefficient, thereby providing a phase shift of 45 degrees. By changing the weights or gains of each input, different phase shifts can be achieved. FIG. 5I depicts an example of a fine phase shifter, where each of the gains are separately controllable (with a 0.707 coefficient depicted for each gain in this example) to achieve a desired phase shift.

(147) FIG. 5J is a schematic diagram of one embodiment of a polyphase filter **410** for a polyphase transmit system. The polyphase filter **410** is a resistor-capacitor (RC) polyphase filter, although other types of polyphase filters, such as inductor-capacitor (LC) polyphase filters, can be used in a polyphase transmit system.

(148) As shown in FIG. 5J, the polyphase filter **410** includes a first resistor **401** between an I input and an I output, a second resistor **402** between a Q input and a Q output, a third resistor **403** between an I_B input and an I_B output, and a fourth resistor **404** between a Q_B input and a Q_B output. The polyphase filter **410** further includes a first capacitor **405** between the I input and Q_B output, a second capacitor **406** between the Q input and the I output, a third capacitor **407** between the I_B input and the Q output, and a fourth capacitor **408** between the Q_B input and the I_B output.

(149) The polyphase filter **410** has a corner frequency F_c .

(150) FIG. 5K is one example of gain versus frequency for a polyphase filter. The polyphase filter provides a notch at frequency F_c , in this example. The notch frequency can be set to attenuate noise or positioned in the location of other frequency bands used in the device (for example, in a mobile phone) to reduce transmit interference.

(151) FIG. 5L is one example of an in-phase/quadrature-phase (I/Q) diagram for frequency shifting in a polyphase transmit system. The frequency shifting can be achieved by a polyphase filter, such as those shown in FIGS. 5B and 5K. A frequency shift of F can be provided as a rotating I/Q vectors, with $+F$ achieved for I leading Q and $-F$ achieved for Q leading I. By changing the capacitor connections, such a change in the filtering characteristics can be achieved.

(152) FIG. 5M is a schematic diagram of one embodiment of a polyphase transmit system **520** using multiple antenna polarizations. In this example, a patch antenna **515** receives a horizontally-polarized non-inverted transmit signal H, a horizontally-polarized inverted transmit signal H_B, a vertically-polarized non-inverted transmit signal V, and a vertically-polarized inverted transmit signal V_B.

(153) As shown in FIG. 5M, the polyphase transmit system **520** further includes first power amplifiers **501a**, **501b**, **501c**, **501d**, second power amplifiers **502a**, **502b**, **502c**, **502d**, combiners **503**, **504**, **505**, **506**, and termination resistors **511**, **512**, **513**, **514**. The power amplifiers process four vertically polarized signals VI (0°), VQ (90°), VI_B (180°), and VQ_B (270°), and four horizontally polarized signals HI (0°), HQ (90°), HI_B (180°), and HQ_B (270°) as shown.

(154) Any of the polyphase transmit signals herein can use multiple antenna polarizations and/or differentially-driven patch antennas.

(155) FIG. 6A is a perspective view of one embodiment of a module **140** that operates with beamforming. FIG. 6B is a cross-section of the module **140** of FIG. 6A taken along the lines 6B-6B.

(156) The module **140** includes a laminated substrate or laminate **141**, a semiconductor die or IC **142**, surface mount components **143**, and an antenna array including patch antenna elements **151-166**.

(157) Although one embodiment of a module is shown in FIGS. 6A and 6B, the teachings herein are applicable to modules implemented in a wide variety of ways. For example, a module can include a different arrangement of and/or number of antenna elements, dies, and/or surface mount

components. Additionally, the module **140** can include additional structures and components including, but not limited to, encapsulation structures, shielding structures, and/or wirebonds.

(158) In the illustrated embodiment, the antenna elements **151-166** are formed on a first surface of the laminate **141**, and can be used to transmit and/or receive signals. Although the illustrated antenna elements **151-166** are rectangular, the antenna elements **151-166** can be shaped in other ways. Additionally, although a 4×4 array of antenna elements is shown, more or fewer antenna elements can be provided. Moreover, antenna elements can be arrayed in other patterns or configurations. Furthermore, in another embodiment, multiple antenna arrays are provided, such as separate antenna arrays for transmit and receive and/or multiple antenna arrays for MIMO and/or switched diversity.

(159) In certain implementations, the antenna elements **151-166** are implemented as patch antennas. A patch antenna can include a planar antenna element positioned over a ground plane. A patch antenna can have a relatively thin profile and exhibit robust mechanical strength. In certain configurations, the antenna elements **151-166** are implemented as patch antennas with planar antenna elements formed on the first surface of the laminate **141** and the ground plane formed using an internal conductive layer of the laminate **141**.

(160) Although an example with patch antennas is shown, a module can include any suitable antenna elements, including, but not limited to, patch antennas, dipole antennas, ceramic resonators, stamped metal antennas, and/or laser direct structuring antennas.

(161) In the illustrated embodiment, the IC **142** and the surface mount components **143** are on a second surface of the laminate **141** opposite the first surface.

(162) In certain implementations, the IC **142** includes signal conditioning circuits associated with the antenna elements **151-166**. In one embodiment, the IC **142** includes a serial interface, such as a mobile industry processor interface radio frequency front-end (MIPI RFFE) bus and/or inter-integrated circuit (I2C) bus that receives data for controlling the signal conditioning circuits, such as the amount of phase shifting provided by phase shifters. In another embodiment, the IC **142** includes signal conditioning circuits associated with the antenna elements **151-166** and an integrated transceiver.

(163) The laminate **141** can be implemented in a variety of ways, and can include for example, conductive layers, dielectric layers, solder masks, and/or other structures. The number of layers, layer thicknesses, and materials used to form the layers can be selected based on a wide variety of factors, which can vary with application. The laminate **141** can include vias for providing electrical connections to signal feeds and/or ground feeds of the antenna elements **151-166**. For example, in certain implementations, vias can aid in providing electrical connections between signaling conditioning circuits of the IC **142** and corresponding antenna elements.

(164) The module **140** can be included in a communication system, such as a mobile phone or base station. In one example, the module **140** is attached to a phone board of a mobile phone.

(165) FIG. 7 is a schematic diagram of one embodiment of a mobile device **800**. The mobile device **800** includes a baseband system **801**, a transceiver **802**, a front end system **803**, antennas **804**, a power management system **805**, a memory **806**, a user interface **807**, and a battery **808**.

(166) The mobile device **800** can be used to communicate using a wide variety of communications technologies, including, but not limited to, 2G, 3G, 4G (including LTE, LTE-Advanced, and LTE-Advanced Pro), 5G NR, WLAN (for instance, WiFi), WPAN (for instance, Bluetooth and ZigBee), WMAN (for instance, WiMax), and/or GPS technologies.

(167) The transceiver **802** generates RF signals for transmission and processes incoming RF signals received from the antennas **804**. It will be understood that various functionalities associated with the transmission and receiving of RF signals can be achieved by one or more components that are collectively represented in FIG. 7 as the transceiver **802**. In one example, separate components (for instance, separate circuits or dies) can be provided for handling certain types of RF signals.

(168) The front end system **803** aids in conditioning signals transmitted to and/or received from the

antennas **804**. In the illustrated embodiment, the front end system **803** includes antenna tuning circuitry **810**, power amplifiers (PAs) **811**, low noise amplifiers (LNAs) **812**, filters **813**, switches **814**, and signal splitting/combining circuitry **815**. However, other implementations are possible. (169) For example, the front end system **803** can provide a number of functionalities, including, but not limited to, amplifying signals for transmission, amplifying received signals, filtering signals, switching between different bands, switching between different power modes, switching between transmission and receiving modes, duplexing of signals, multiplexing of signals (for instance, diplexing or triplexing), or some combination thereof.

(170) In certain implementations, the mobile device **800** supports carrier aggregation, thereby providing flexibility to increase peak data rates. Carrier aggregation can be used for both Frequency Division Duplexing (FDD) and Time Division Duplexing (TDD), and may be used to aggregate a plurality of carriers or channels. Carrier aggregation includes contiguous aggregation, in which contiguous carriers within the same operating frequency band are aggregated. Carrier aggregation can also be non-contiguous, and can include carriers separated in frequency within a common band or in different bands.

(171) The antennas **804** can include antennas used for a wide variety of types of communications. For example, the antennas **804** can include antennas for transmitting and/or receiving signals associated with a wide variety of frequencies and communications standards.

(172) In certain implementations, the antennas **804** support MIMO communications and/or switched diversity communications. For example, MIMO communications use multiple antennas for communicating multiple data streams over a single radio frequency channel. MIMO communications benefit from higher signal to noise ratio, improved coding, and/or reduced signal interference due to spatial multiplexing differences of the radio environment. Switched diversity refers to communications in which a particular antenna is selected for operation at a particular time. For example, a switch can be used to select a particular antenna from a group of antennas based on a variety of factors, such as an observed bit error rate and/or a signal strength indicator.

(173) The mobile device **800** can operate with beamforming in certain implementations. For example, the front end system **803** can include amplifiers having controllable gain and phase shifters having controllable phase to provide beam formation and directivity for transmission and/or reception of signals using the antennas **804**. For example, in the context of signal transmission, the amplitude and phases of the transmit signals provided to the antennas **804** are controlled such that radiated signals from the antennas **804** combine using constructive and destructive interference to generate an aggregate transmit signal exhibiting beam-like qualities with more signal strength propagating in a given direction. In the context of signal reception, the amplitude and phases are controlled such that more signal energy is received when the signal is arriving to the antennas **804** from a particular direction. In certain implementations, the antennas **804** include one or more arrays of antenna elements to enhance beamforming.

(174) The baseband system **801** is coupled to the user interface **807** to facilitate processing of various user input and output (I/O), such as voice and data. The baseband system **801** provides the transceiver **802** with digital representations of transmit signals, which the transceiver **802** processes to generate RF signals for transmission. The baseband system **801** also processes digital representations of received signals provided by the transceiver **802**. As shown in FIG. 7, the baseband system **801** is coupled to the memory **806** of facilitate operation of the mobile device **800**.

(175) The memory **806** can be used for a wide variety of purposes, such as storing data and/or instructions to facilitate the operation of the mobile device **800** and/or to provide storage of user information.

(176) The power management system **805** provides a number of power management functions of the mobile device **800**. In certain implementations, the power management system **805** includes a PA supply control circuit that controls the supply voltages of the power amplifiers **811**. For

example, the power management system **805** can be configured to change the supply voltage(s) provided to one or more of the power amplifiers **811** to improve efficiency, such as power added efficiency (PAE).

(177) As shown in FIG. 7, the power management system **805** receives a battery voltage from the battery **808**. The battery **808** can be any suitable battery for use in the mobile device **800**, including, for example, a lithium-ion battery.

(178) FIG. 8 is a schematic diagram of a power amplifier system **860** according to another embodiment. The illustrated power amplifier system **860** includes a baseband processor **841**, a transmitter/observation receiver **842**, a power amplifier (PA) **843**, a directional coupler **844**, front-end circuitry **845**, an antenna **846**, a PA bias control circuit **847**, and a PA supply control circuit **848**. The illustrated transmitter/observation receiver **842** includes an I/Q modulator **857**, a mixer **858**, and an analog-to-digital converter (ADC) **859**. In certain implementations, the transmitter/observation receiver **842** is incorporated into a transceiver.

(179) The baseband processor **841** can be used to generate an in-phase (I) signal and a quadrature-phase (Q) signal, which can be used to represent a sinusoidal wave or signal of a desired amplitude, frequency, and phase. For example, the I signal can be used to represent an in-phase component of the sinusoidal wave and the Q signal can be used to represent a quadrature-phase component of the sinusoidal wave, which can be an equivalent representation of the sinusoidal wave. In certain implementations, the I and Q signals can be provided to the I/Q modulator **857** in a digital format. The baseband processor **841** can be any suitable processor configured to process a baseband signal. For instance, the baseband processor **841** can include a digital signal processor, a microprocessor, a programmable core, or any combination thereof. Moreover, in some implementations, two or more baseband processors **841** can be included in the power amplifier system **860**.

(180) The I/Q modulator **857** can be configured to receive the I and Q signals from the baseband processor **841** and to process the I and Q signals to generate an RF signal. For example, the I/Q modulator **857** can include digital-to-analog converters (DACs) configured to convert the I and Q signals into an analog format, mixers for upconverting the I and Q signals to RF, and a signal combiner for combining the upconverted I and Q signals into an RF signal suitable for amplification by the power amplifier **843**. In certain implementations, the I/Q modulator **857** can include one or more filters configured to filter frequency content of signals processed therein.

(181) The power amplifier **843** can receive the RF signal from the I/Q modulator **857**, and when enabled can provide an amplified RF signal to the antenna **846** via the front-end circuitry **845**.

(182) The front-end circuitry **845** can be implemented in a wide variety of ways. In one example, the front-end circuitry **845** includes one or more switches, filters, duplexers, multiplexers, and/or other components. In another example, the front-end circuitry **845** is omitted in favor of the power amplifier **843** providing the amplified RF signal directly to the antenna **846**.

(183) The directional coupler **844** senses an output signal of the power amplifier **823**. Additionally, the sensed output signal from the directional coupler **844** is provided to the mixer **858**, which multiplies the sensed output signal by a reference signal of a controlled frequency. The mixer **858** operates to generate a downshifted signal by downshifting the sensed output signal's frequency content. The downshifted signal can be provided to the ADC **859**, which can convert the downshifted signal to a digital format suitable for processing by the baseband processor **841**.

Including a feedback path from the output of the power amplifier **843** to the baseband processor **841** can provide a number of advantages. For example, implementing the baseband processor **841** in this manner can aid in providing power control, compensating for transmitter impairments, and/or in performing digital pre-distortion (DPD). Although one example of a sensing path for a power amplifier is shown, other implementations are possible.

(184) The PA supply control circuit **848** receives a power control signal from the baseband processor **841**, and controls supply voltages of the power amplifier **843**. In the illustrated configuration, the PA supply control circuit **848** generates a first supply voltage V_{sub_CC1} for

powering an input stage of the power amplifier **843** and a second supply voltage V.sub.CC2 for powering an output stage of the power amplifier **843**. The PA supply control circuit **848** can control the voltage level of the first supply voltage V.sub.CC1 and/or the second supply voltage V.sub.CC2 to enhance the power amplifier system's PAE.

(185) The PA supply control circuit **848** can employ various power management techniques to change the voltage level of one or more of the supply voltages over time to improve the power amplifier's power added efficiency (PAE), thereby reducing power dissipation.

(186) One technique for improving efficiency of a power amplifier is average power tracking (APT), in which a DC-to-DC converter is used to generate a supply voltage for a power amplifier based on the power amplifier's average output power. Another technique for improving efficiency of a power amplifier is envelope tracking (ET), in which a supply voltage of the power amplifier is controlled in relation to the envelope of the RF signal. Thus, when a voltage level of the envelope of the RF signal increases the voltage level of the power amplifier's supply voltage can be increased. Likewise, when the voltage level of the envelope of the RF signal decreases the voltage level of the power amplifier's supply voltage can be decreased to reduce power consumption.

(187) In certain configurations, the PA supply control circuit **848** is a multi-mode supply control circuit that can operate in multiple supply control modes including an APT mode and an ET mode. For example, the power control signal from the baseband processor **841** can instruct the PA supply control circuit **848** to operate in a particular supply control mode.

(188) As shown in FIG. **8**, the PA bias control circuit **847** receives a bias control signal from the baseband processor **841**, and generates bias control signals for the power amplifier **843**. In the illustrated configuration, the bias control circuit **847** generates bias control signals for both an input stage of the power amplifier **843** and an output stage of the power amplifier **843**. However, other implementations are possible.

(189) Applications

(190) Some of the embodiments described above have provided examples in connection with wireless devices or mobile phones. However, the principles and advantages of the embodiments can be used for any other suitable systems or apparatus.

(191) For example, the embodiments herein can be implemented in various electronic devices. Examples of the electronic devices can include, but are not limited to, consumer electronic products, parts of the consumer electronic products, electronic test equipment, etc. Examples of the electronic devices can also include, but are not limited to, memory chips, memory modules, circuits of optical networks or other communication networks, and disk driver circuits. The consumer electronic products can include, but are not limited to, a mobile phone, a telephone, a television, a computer monitor, a computer, a hand-held computer, a personal digital assistant (PDA), a microwave, a refrigerator, an automobile, a stereo system, a cassette recorder or player, a DVD player, a CD player, a VCR, an MP3 player, a radio, a camcorder, a camera, a digital camera, a portable memory chip, a washer, a dryer, a washer/dryer, a copier, a facsimile machine, a scanner, a multi-functional peripheral device, a wrist watch, a clock, etc. Further, the electronic devices can include unfinished products.

CONCLUSION

(192) Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise,” “comprising,” and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to.” The word “coupled”, as generally used herein, refers to two or more elements that may be either directly connected, or connected by way of one or more intermediate elements. Likewise, the word “connected”, as generally used herein, refers to two or more elements that may be either directly connected, or connected by way of one or more intermediate elements. Additionally, the words “herein,” “above,” “below,” and words of similar import, when used in this application, shall refer to this application as a whole and not to any particular portions of this application. Where the

context permits, words in the above Detailed Description using the singular or plural number may also include the plural or singular number respectively. The word “or” in reference to a list of two or more items, that word covers all of the following interpretations of the word: any of the items in the list, all of the items in the list, and any combination of the items in the list.

(193) Moreover, conditional language used herein, such as, among others, “may,” “could,” “might,” “can,” “e.g.,” “for example,” “such as” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or states. Thus, such conditional language is not generally intended to imply that features, elements and/or states are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or states are included or are to be performed in any particular embodiment.

(194) The above detailed description of embodiments of the invention is not intended to be exhaustive or to limit the invention to the precise form disclosed above. While specific embodiments of, and examples for, the invention are described above for illustrative purposes, various equivalent modifications are possible within the scope of the invention, as those skilled in the relevant art will recognize. For example, while processes or blocks are presented in a given order, alternative embodiments may perform routines having steps, or employ systems having blocks, in a different order, and some processes or blocks may be deleted, moved, added, subdivided, combined, and/or modified. Each of these processes or blocks may be implemented in a variety of different ways. Also, while processes or blocks are at times shown as being performed in series, these processes or blocks may instead be performed in parallel, or may be performed at different times.

(195) The teachings of the invention provided herein can be applied to other systems, not necessarily the system described above. The elements and acts of the various embodiments described above can be combined to provide further embodiments.

(196) While certain embodiments of the inventions have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosure. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the methods and systems described herein may be made without departing from the spirit of the disclosure. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure.

Claims

1. A mobile device comprising: a baseband processor configured to generate digital transmit data representing a transmit signal; an intermediate frequency transceiver including a plurality of digital-to-analog converters configured to convert the digital transmit data into a plurality of analog baseband transmit signals of different phases, and a first complex mixer configured to generate a plurality of intermediate frequency transmit signals of different phases based on the plurality of analog baseband transmit signals; a front-end system including a second complex mixer configured to generate a plurality of radio frequency transmit signals of different phases based on the plurality of intermediate frequency transmit signals, and a polyphase power amplifier configured to receive the plurality of radio frequency transmit signals and to output an amplified radio frequency signal; and an antenna configured to transmit the amplified radio frequency signal.
2. The radio frequency transit system of claim 1 wherein the polyphase power amplifier provides insensitivity to a variation in an impedance of the antenna.
3. The radio frequency transmit system of claim 2 wherein the variation is a change in voltage

standing wave ratio (VSWR).

4. The radio frequency transit system of claim 1 wherein the plurality of intermediate frequency transmit signals includes an intermediate frequency in-phase transmit signal and an intermediate frequency quadrature-phase transmit signal, and the plurality of radio frequency transmit signals includes a radio frequency in-phase transmit signal and a radio frequency quadrature-phase transmit signal.

5. The radio frequency transit system of claim 4 wherein the front-end system further includes a coarse phase shifter configured to selectively provide a ninety degree phase shift to the radio frequency in-phase transmit signal and the radio frequency quadrature-phase transmit signal.

6. The radio frequency transit system of claim 5 wherein the front-end system further includes a coarse phase shifter configured to provide a fine phase shift to the radio frequency in-phase transmit signal and the radio frequency quadrature-phase transmit signal.

7. The radio frequency transit system of claim 4 wherein the front-end system further includes a polyphase filter configured to filter the radio frequency in-phase transmit signal and the radio frequency quadrature-phase transmit signal.

8. The radio frequency transit system of claim 4 wherein the polyphase amplifier includes a first controllable gain power amplifier configured to receive the radio frequency in-phase transmit signal, a second controllable gain power amplifier configured to receive the radio frequency quadrature-phase transmit signal, and a combiner configured to generate the amplified radio frequency signal by combining an output of the first controllable gain power amplifier and an output of the second controllable gain power amplifier.

9. The radio frequency transit system of claim 1 wherein the antenna is a patch antenna.

10. The radio frequency transit system of claim 1 wherein the intermediate frequency transceiver includes a first local oscillator configured to provide the first complex mixer with a first plurality of local oscillator signals, and the front end system further includes a second local oscillator configured to provide the second complex mixer with a second plurality of local oscillator signals.

11. A polyphase transmit system comprising: an intermediate frequency transceiver including a first complex mixer configured to output a plurality of intermediate frequency transmit signals of different phases; an intermediate frequency to radio frequency module including a second complex mixer configured to generate a plurality of radio frequency transmit signals of different phases based on the plurality of intermediate frequency transmit signals, and a polyphase power amplifier configured to receive the plurality of radio frequency transmit signals and to output an amplified radio frequency signal; and an antenna configured to transmit the amplified radio frequency signal.

12. The polyphase transmit system of claim 11 wherein the polyphase power amplifier provides insensitivity to a variation in an impedance of the antenna.

13. The polyphase transmit system of claim 11 wherein the plurality of intermediate frequency transmit signals includes an intermediate frequency in-phase transmit signal and an intermediate frequency quadrature-phase transmit signal, and the plurality of radio frequency transmit signals includes a radio frequency in-phase transmit signal and a radio frequency quadrature-phase transmit signal.

14. The polyphase transmit system of claim 13 wherein the intermediate frequency to radio frequency module further includes a coarse phase shifter configured to selectively provide a ninety degree phase shift to the radio frequency in-phase transmit signal and the radio frequency quadrature-phase transmit signal.

15. The polyphase transmit system of claim 14 wherein the intermediate frequency to radio frequency module further includes a coarse phase shifter configured to provide a fine phase shift to the radio frequency in-phase transmit signal and the radio frequency quadrature-phase transmit signal.

16. The polyphase transmit system of claim 13 wherein the intermediate frequency to radio frequency module further includes a polyphase filter configured to filter the radio frequency in-

phase transmit signal and the radio frequency quadrature-phase transmit signal.

17. The polyphase transmit system of claim 13 wherein the polyphase amplifier includes a first controllable gain power amplifier configured to receive the radio frequency in-phase transmit signal, a second controllable gain power amplifier configured to receive the radio frequency quadrature-phase transmit signal, and a combiner configured to combine an output of the first controllable gain power amplifier and an output of the second controllable gain power amplifier.

18. The polyphase transmit system of claim 11 wherein the antenna is a patch antenna.

19. The polyphase transmit system of claim 11 wherein the intermediate frequency transceiver includes a first local oscillator configured to provide the first complex mixer with a first plurality of local oscillator signals, and the intermediate frequency to radio frequency module further includes a second local oscillator configured to provide the second complex mixer with a second plurality of local oscillator signals.

20. A method of transmitting in a mobile device, the method system comprising: outputting a plurality of intermediate frequency transmit signals of different phases from a first complex mixer of an intermediate frequency transceiver; generating a plurality of radio frequency transmit signals of different phases based on the plurality of intermediate frequency transmit signals using a second complex mixer of an intermediate frequency to radio frequency module; amplifying the plurality of radio frequency transmit signals to generate an amplified radio frequency signal using a polyphase power amplifier; and transmitting the amplified radio frequency signal using an antenna.
